

Francesco ROSA



PROFILE SUMMARY

Computer engineer specialized in Artificial Intelligence, Robotics, and Computer Vision, with experience in developing intelligent robotic systems based on deep learning and human-robot collaboration.

CONTACT DETAILS

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<https://ciccio42.github.io/>

PERSONAL INFORMATION

Date of birth: 08-09-1997

Address: Via Grotta Parrella, 16/C
Boscoreale (NA), Italy

SKILLS

- Robotics, Deep Learning, Computer Vision
- Python, C/C++, CUDA
- PyTorch, TensorFlow
- Huggin Face
- ROS/ROS2
- Linux, Git, Docker
- Nvidia Jetson

EDUCATION

PH.D. IN COMPUTER ENGINEERING from *University of Salerno*.

Grade: Excellent.

Nov. 2021– Feb. 2025.

Thesis: “*Video-Conditioned Multi-Task Imitation Learning for Robotic Systems: Enhancing Robustness Through Object-Centric Reasoning*”

- Research in design and development of an innovative Deep Learning-based architecture for controlling robotic manipulators capable of learning and reproducing complex tasks from video demonstrations.

MASTER DEGREE IN COMPUTER ENGINEERING from *University of Salerno*.

Grade: 110/110 Cum Laude

Sept. 2019 - Oct. 2021

Thesis: “*3D Deep Neural Network for the 6DoF Grasp Pose Estimation on an Embedded Platform*”

- Development of a 3D deep neural network for grasp pose estimation, deployable on an embedded platform

BACHELOR DEGREE IN COMPUTER ENGINEERING from *University of Salerno*.

Grade: 110/110 Cum Laude

Sept. 2016 - Oct. 2019

Thesis: “*Creation of a system for managing an automatic irrigation system for a smart pot and integration with an iOS application*”

- Development of a control system for the automatic irrigation of a smart pot. Integration with an iOS application for remote monitoring and management

EXPERIENCE

RESEARCHER at *University of Salerno*.

Nov. 2024–Present

◇ Research and development of data-driven control architectures for **anthropomorphic robotic manipulators**. Use of **Vision-Language-Action** models to make robotic behavior more adaptive and intelligent.

ACADEMIC TUTOR at *University of Salerno*.

Nov. 2021–Jan. 2025

◇ During my Ph.D., I conducted exercise lessons for undergraduate courses, including Fondamenti di Programmazione, Algoritmi e Strutture Dati, Calcolatori Elettronici, Fisica 2, and Fisica Generale.

VISITING PH.D. at *Groupe de Recherche en Informatique, Image, Automatique et Instrumentation de Caen (GREYC)*.

Apr. 2024–Jul. 2024

◇ Conducted research under the supervision of Professor *Luc Brun* on applying GNNs in Robotics Learning. Focused on using GNNs as heuristic estimators for Task Planning. Evaluated state-of-the-art GNNs to assess their effectiveness in heuristic estimation for robotic planning problems.

RESEARCH AND DEVELOPMENT at *Kalya*.

Jan. 2020–Dec. 2021

◇ Innovative start-up in Agriculture 4.0. Design and implementation of the image acquisition and processing system for the intelligent trap “*Sentinella*”. Development of a network of smart traps to estimate pest populations in crops

RESEARCH PROJECT

RESEARCHER FOR EUROPEAN PROJECT H2020 “*FLEXIBLE ASSEMBLY MANUFACTURING WITH HUMAN-ROBOT COLLABORATION AND DIGITAL TWIN MODELS*” at *University of Salerno*.

◇ Development of a real-time Voice-Command and Gesture-Command Recognition system on an Nvidia Jetson Xavier embedded platform installed on a mobile manipulator robot

HONORS AND CERTIFICATIONS

NATIONAL INNOVATION AWARD 2020 – PNI 2020

◇ Finalists at PNI 2020, CleanTech & Energy category, with the start-up Kalya and its project Sentinella

SECOND PLACE START-CUP CAMPANIA

◇ Kalya, with project Sentinella, achieved second place at StartCup Campania 2020

EXCELLENT STUDENT

◇ DIEM Award for Academic Excellence – Bachelor’s and Master’s degrees earned with highest honors (110 cum laude) in 36 and 26 months, respectively